

SHADOWFOX

DATA ANALYST INTERNSHIP

TASK 1 — BEGINNER LEVEL

RETAIL SALES ANALYSIS DASHBOARD

Internship Project Report

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Course:

B.Tech – Computer Science and Engineering (Data Science)

Year:

3rd Year

Institution:

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Academic Year:

2026

ShadowFox Program ID (SF-PID): SF-PID-2026-W6061

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RETAIL SALES ANALYSIS DASHBOARD

1. Project Overview

The **Retail Sales Analysis Dashboard** is a beginner-level data analytics project created to analyze retail sales data and identify useful business patterns.

The project uses sales information such as order date, product, category, region, quantity, sales, cost, and profit. The dataset was cleaned and organized before performing analysis using pivot tables and charts.

The final result is an interactive spreadsheet dashboard that presents important business metrics and visual insights about sales and profit performance.

2. Objectives

The main objectives of this project are:

- To understand and clean a retail sales dataset.
- To identify and remove duplicate records.
- To handle missing values in the dataset.
- To analyze sales based on month, category, region, and product.
- To calculate important business metrics such as total sales, total profit, total orders, and total quantity.
- To analyze monthly sales and profit trends.
- To create charts for better visualization of the data.
- To develop a simple business dashboard.
- To identify useful business insights and provide recommendations.

3. Dataset Description

The dataset contains retail sales information for different orders and products.

The major columns used in the dataset are:

Column	Description
Order ID	Unique identification number for an order
Order Date	Date on which the order was placed
Product	Name of the product
Category	Product category
Region	Geographic sales region
Customer	Customer information
Quantity	Number of products purchased
Unit Price	Price of one unit
Sales	Total sales amount
Cost	Cost associated with the sale
Profit	Profit generated from the sale
Month	Month extracted from the order date

The dataset contains different product categories such as **Electronics, Furniture, Office Supplies, and Accessories**.

The regions included are **North, South, East, West**, along with **Unknown** for records where the region information was missing.

4. Data Cleaning

Data cleaning was performed before starting the analysis.

4.1 Removing Duplicate Records

Duplicate records were identified and removed from the dataset to prevent the same transaction from being counted more than once.

Three duplicate records were removed during the cleaning process.

4.2 Handling Missing Values

Some records contained missing values in fields such as Product, Region, and Customer.

Instead of deleting these records, the missing values were replaced with "**Unknown**" so that useful transaction information was not lost.

The Unknown region was kept separately in the analysis to maintain transparency about the original data.

4.3 Date Standardization

The Order Date column was checked and standardized so that the dates could be used correctly for monthly analysis.

4.4 Creating the Month Column

A separate Month column was created from the Order Date.

This helped in analyzing monthly sales and profit trends.

The formula used was:

```
=TEXT(B2, "mmm-yyyy")
```

5. Analysis Methodology

The analysis was performed using **Google Sheets**.

Pivot tables were created to summarize the cleaned dataset.

The following analyses were performed:

5.1 Monthly Sales Analysis

A pivot table was created using Month and Sales to identify how sales changed over time.

A line chart was then created to visualize the monthly sales trend.

5.2 Category Sales Analysis

Sales were grouped according to product category.

This helped identify which product category generated the highest sales.

5.3 Regional Sales Analysis

Sales were grouped by region to compare the performance of different geographic regions.

5.4 Product Sales Analysis

Products were ranked based on their sales amount.

A Top 10 Products chart was created to identify the products contributing the most to sales.

5.5 Monthly Profit Analysis

Profit was grouped by month to understand changes in profitability over time.

A line chart was created to visualize the monthly profit trend.

6. Key Performance Metrics

The dashboard contains the following important KPIs:

- **Total Sales**
- **Total Profit**
- **Total Orders**
- **Total Quantity**
- **Average Order Value**

These metrics provide a quick overview of the overall business performance.

The KPI values were linked from the Analysis sheet to the Dashboard so that the dashboard could display the calculated results automatically.

7. Dashboard

A simple **Retail Sales Analysis Dashboard** was created using Google Sheets.

The dashboard contains five main charts:

1. **Monthly Sales Trend** – shows changes in sales over different months.
2. **Sales by Category** – compares sales generated by different product categories.
3. **Sales by Region** – compares sales performance across regions.
4. **Top 10 Products by Sales** – identifies the products with the highest sales.
5. **Monthly Profit Trend** – shows changes in profit over time.

The dashboard also contains KPI cards for Total Sales, Total Profit, Total Orders, Total Quantity, and Average Order Value.

8. Key Findings

Based on the analysis performed on the dataset, the following findings were identified:

Finding 1: Best-Performing Category

Accessories generated the highest sales among the product categories analyzed.

This indicates that Accessories had strong customer demand and contributed significantly to overall sales.

Finding 2: Best-Performing Region

The **South region** recorded the highest sales among the identified regions.

This indicates that the South region was the strongest-performing geographic market in the dataset.

Finding 3: Highest-Sales Month

April 2025 recorded the highest monthly sales.

This indicates that sales reached their peak during this period.

Finding 4: Top-Selling Product

Monitor was identified as the top-selling product based on sales.

This suggests that monitors were an important contributor to the company's overall sales.

Finding 5: Lowest-Performing Region

Among the identified regions, **East** recorded the lowest sales.

Some records contained missing region information and were categorized as **Unknown** during data cleaning. Therefore, Unknown was not treated as an actual geographic region when identifying the lowest-performing region.

9. Business Recommendations

Based on the findings, the following recommendations can be made:

9.1 Focus on Accessories

Since Accessories generated the highest sales, the business can maintain sufficient inventory and consider promoting related accessories.

9.2 Study the South Region

The South region performed strongly. The business can analyze the factors behind this performance and use successful strategies from this region in other areas.

9.3 Analyze April Sales

Since April 2025 had the highest sales, the business can investigate whether promotions, seasonal demand, discounts, or other factors contributed to the increase.

9.4 Maintain Monitor Availability

Since Monitor was the top-selling product, maintaining sufficient stock can help avoid missed sales opportunities.

9.5 Improve East Region Performance

Since East had the lowest sales among the identified regions, the business can investigate customer demand, pricing, product availability, and marketing activities in this region.

9.6 Improve Data Quality

Missing region and other customer/product information should be captured more consistently in future transactions to improve the accuracy of business analysis.

10. Tools and Technologies Used

The following tools and techniques were used:

- **Google Sheets** – data cleaning, analysis, pivot tables, charts, and dashboard creation
- **Spreadsheet Formulas** – calculations and month extraction
- **Pivot Tables** – data summarization and comparison
- **Charts** – visual representation of trends and comparisons
- **Dashboard Design** – presentation of KPIs and business insights

11. Conclusion

The Retail Sales Analysis Dashboard demonstrates how raw sales data can be transformed into useful business information through data cleaning, analysis, visualization, and dashboard creation.

The project helped identify important sales patterns, including the strongest product category, best-performing region, highest-sales month, top-selling product, and lowest-performing region.

The dashboard provides a simple way for business users to understand sales and profit performance without manually analyzing the complete dataset.

This project also provided practical experience in important data analyst activities such as data cleaning, pivot table analysis, KPI creation, data visualization, dashboard development, and business insight generation.